THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(d)** 3 x 10^5	2 x 10^30 / 6 x 10^24.
3	**(b)** the theory is strongly supported but remains open to tests	Science is provisional.
4	**(d)** A is false but R is true.	A is false (theories can be refined); R is true.
5	**(a)** 5.4 cm	Subtract the zero error.
6	**(d)** dosage of a medicine for a child	Safety critical.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(b)** 1 per cent	0.2/20.4.
9	**(d)** check the experiment and the assumptions in the model	Investigate.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area = 96 square cm (1); About 2 per cent (1).	1+1
12	A straight line shows proportionality (1); Easy to read the slope and spot errors (1).	1+1
13	Distance = 3 x 10^5 x 3.15 x 10^7 (1); = 9.45 x 10^12 km (1).	1+1
14	Mean = 9.76 m/s^2 (1); Range = 0.3 m/s^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mean = 42.6/3 (1); = 14.2 cm (1); Greatest deviation = 0.1 cm (1).	1+1+1
16	Shows how reliable the result is (1); e.g. 12.3 +/- 0.1 cm (1); Allows comparison with other results (1).	1+1+1
17	Rising curve (1); Not a straight line (1); Longer pendulum has longer period (1).	1+1+1
	OR: Mean = 54.4 s for 30 (1); T = 1.81 s (1); Spread about 0.4 / 54.4 = 0.7 per cent (1).	1+1+1
18	Volume = 144 cubic m (1); Mass = 144 x 1.2 (1); = about 173 kg (1).	1+1+1
	OR: Informed consent (1); Safety tests first (1); Honest reporting of side effects (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Apparatus: thread, bob, stand, stopwatch, scale (1); Independent: length; dependent: time period (1); Constants: mass, small angle (1); Time 20 oscillations and repeat (1); Plot T against L and conclude that T increases with L (1).	1+1+1+1+1
	OR: Same mass and volume (1); Vary only temperature (1); Record times (1); Graph of time vs temperature (1); Errors: reaction time and stirring (1).	1+1+1+1+1
20	Strength: evidence-based (1); Strength: self-correcting (1); Limitation: cannot test untestable ideas (1); Chance: penicillin discovery (1); Persistence: many experiments repeated (1).	1+1+1+1+1
	OR: Observations of planets (1); Copernicus's model (1); Galileo's telescope (1); Kepler's laws (1); Debate and acceptance (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 15 per cent	3 x 5 per cent.
2	**(b)** to test the procedure and find problems	Trial run.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(d)** 4.5 x 10^-4	Standard form.
5	**(b)** precise but not accurate	Close to each other but away from true.
6	**(d)** A is false but R is true.	A false; R true.
7	**(d)** 0.5 m	Uncertainty must exceed the difference.
8	**(a)** accurate on average but not precise	Mean close to the true value.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(c)** increase the number of trials and use a control	More data.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Millions of data points and equations (1); Many calculations are needed quickly (1).	1+1
12	Total energy must remain constant (1); A violation signals an error in the model (1).	1+1
13	A future observation may contradict it (1); Science always remains open to revision (1).	1+1
14	Zero error shifts all readings in the same direction (1); Reaction time varies unpredictably (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Must be possible to show it false (1); Example of testable: heating increases the rate of dissolving (1); Not falsifiable: unseen forces cause everything (1).	1+1+1
16	1 litre = 1000 mL (1); 1000/0.05 (1); = 20000 drops (1).	1+1+1
17	May be due to chance or error (1); Reproducibility is the test of reliability (1); Needs further study (1).	1+1+1
	OR: Error = 0.02 (1); 0.02/2.70 x 100 = 0.74 per cent (1); Careful measurement of mass and volume (1).	1+1+1
18	Units of distance x time are m s (1); Speed has unit m/s (1); Unit mismatch shows error (1).	1+1+1
	OR: Melting point depends slightly on pressure (1); Condition must be clear for repeating (1); Conditions are part of a law (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Volume = 40 cubic cm (1); Density = 3.39 g/cm3 (1); Uncertainty: 2 + 2.5 + 5 + 0.1 about 9.6 per cent (1); Range about 3.1 to 3.7 g/cm3 (1); Possibly a light alloy or a mineral; needs better measurement (1).	1+1+1+1+1
	OR: Moles = 0.05/18 = 2.8 x 10^-3 (1); Molecules = 2.8 x 10^-3 x 6 x 10^23 = 1.7 x 10^21 (1); Atoms = 3 x that = 5 x 10^21 (1); Shows how small atoms are (1); Estimation gives order of magnitude (1).	1+1+1+1+1
20	Standard units help sharing data (1); Mars Climate Orbiter used mixed units (1); Spacecraft was lost (1); Importance of checking units (1); Open data and cooperation (1).	1+1+1+1+1
	OR: Same mass of water and same container (1); Same mass of each fuel burnt (1); Same starting temperature and distance of flame (1); Repeat and average (1); Errors: heat loss, draught, reading errors (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** the gas is at low pressure	Large separation.
2	**(d)** A is false but R is true.	A is false because a hypothesis is a proposed explanation; R is true.
3	**(d)** 1.08 x 10^9	3 x 10^5 x 3600 km/h.
4	**(b)** 25000	2.5 x 10000.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(d)** 0.3 per cent	0.03/9.81 x 100.
7	**(d)** 12.34 g	Matches least count.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(c)** 10^-4 m	10^6 x 10^-10.
10	**(b)** a control group without warm water	Controlled trial.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Theories are well-tested explanations (1); They are not just guesses (1).	1+1
12	5.4 x 10^4 x 1000 / 3600 (1); = 15000 m/s (1).	1+1
13	Experts check methods and conclusions (1); Mistakes are corrected before acceptance (1).	1+1
14	Two things occurring together need not be linked as cause and effect (1); e.g. ice cream sales and drowning rise in summer (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Density = 2.7 g/cm3 (1); = 2700 kg/m3 (1); Aluminium (1).	1+1+1
16	Tests the method on a small scale (1); Finds problems early (1); e.g. trial run of a dissolving experiment (1).	1+1+1
17	Compare methods and calibration (1); Repeat with the same procedure (1); Seek independent confirmation (1).	1+1+1
	OR: Control plants without tonic (1); Same soil and water (1); Compare growth (1).	1+1+1
18	Small-angle assumption breaks down (1); Period depends on amplitude then (1); Hence the model needs revision (1).	1+1+1
	OR: Saha: stellar ionisation (1); Raman: scattering of light (1); Their work used careful experiments and reasoning (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Solar system: Copernicus to Kepler (1); Atom: Dalton to Bohr (1); Cell: Hooke to modern (1); Evidence from instruments (1); Models revised (1).	1+1+1+1+1
	OR: Replication tests results (1); Peer review checks methods (1); Transparency shares data (1); Example: retracted papers (1); Self-correcting science (1).	1+1+1+1+1
20	Check other factors like irrigation (1); Control experiments (1); Statistical analysis (1); Replicate across regions (1); Correlation does not prove causation (1).	1+1+1+1+1
	OR: Speed = distance/time (1); Direct: distance vs time at constant speed (1); Inverse: time vs speed at fixed distance (1); Graphs show relations (1); Units check formulas (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.